Trade Tariffs and Exchange Rates in a Multilateral World

Theoretical derivations

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1 Introduction

In a two-country model, an import tariff appreciates the tariffing country's currency under the Marshall–Lerner condition. In a multilateral setting, a discriminatory tariff also reallocates expenditure across foreign suppliers, so bilateral exchange-rate responses need not share a sign. These notes characterize the exchange-rate responses in an N-country model. The general equilibrium response is $(-J)^{-1}F$, where F is the vector of trade-balance disturbances created by the tariff at unchanged exchange rates and J is the exchange-rate adjustment mechanism. Under symmetry and a multilateral Marshall–Lerner condition, the tariffing country always appreciates against the target, but the sign of its bilateral exchange-rate response against an untariffed bystander is ambiguous and depends on whether the cross-origin substitution elasticity exceeds a threshold $N\rho_1^*$. Despite this bilateral sign ambiguity, the tariffing country's nominal effective exchange rate appreciates independently of the cross-origin elasticity and N. The exact cancellation behind this aggregate result and the linear threshold are properties of the symmetric nested-CES model, but the disturbance-adjustment decomposition, and the distinction between aggregate home-versus-foreign adjustment and reallocation across foreign suppliers, are more general.

2 N-Country Model and Equilibrium

2.1 Production and preferences

Countries are indexed by $i, j \in \{1, \ldots, N\}$; generically, A denotes the tariff imposer, B the tariff target, and C an untariffed bystander. Each country produces a country-specific tradable and a non-tradable from labor:

$$Y_{T_i} = A_{T_i} L_{T_i}, \qquad Y_{N_i} = A_{N_i} L_{N_i}, \qquad L_{T_i} + L_{N_i} = L_i, \tag{1}$$

with perfect competition and producer-price normalization $P_{T_i} = 1$ in i's currency, so

$$w_i = A_{T_i}, \qquad P_{N_i} = A_{T_i}/A_{N_i}. \tag{2}$$

Wages are constant in own currency, so all relative adjustment runs through exchange rates. Preferences are nested CES with share-linear weights:

$$C_i = C_{T_i}^{\alpha_T}\, C_{N_i}^{1-\alpha_T} \qquad \text{(tradable share } \alpha_T) \tag{3}$$

$$C_{T_i} = \left[\alpha_D^{1/\eta}\, C_{T_i i}^{\frac{\eta-1}{\eta}} + (1-\alpha_D)^{1/\eta}\, M_i^{\frac{\eta-1}{\eta}}\right]^{\frac{\eta}{\eta-1}} \qquad \text{(home weight } \alpha_D,\ \text{elasticity } \eta) \tag{4}$$

$$M_i = \left[\, \textstyle\sum_{j\neq i} \beta_{ij}^{1/\rho}\, C_{T_j i}^{\frac{\rho-1}{\rho}}\right]^{\frac{\rho}{\rho-1}} \qquad \text{(origin weights } \beta_{ij},\ \text{elasticity } \rho). \tag{5}$$

η governs substitution between domestic and imported tradables (the home-versus-foreign tradables margin); ρ governs substitution across foreign origins conditional on importing (the foreign-supplier tradables reallocation margin, inactive at $N = 2$ and active at $N \geq 3$).

The *symmetric benchmark* imposes $\beta_{ij} = 1/(N-1)$, common parameters $(\alpha_T, \alpha_D, \eta, \rho)$, and equal endowments and productivities. Home bias α_D keeps home and foreign tradables asymmetric; full home-versus-foreign tradables symmetry would additionally require $\rho = \eta$ and $\alpha_D = 1/N$, and is not imposed.

2.2 Tariffs, prices, income, and trade balance

Tariffs $\tau_{ij} \geq 0$ are ad valorem, levied by i on imports from j, with $\tau_{ii} = 0$. Consumer prices in i's currency are given by $p_{ij} = (1+\tau_{ij})\, e_{ij}$, $p_{ii} = 1$, and price indices are given by:

$$P_{M_i} = \left[\textstyle\sum_{j\neq i} \beta_{ij}\, p_{ij}^{1-\rho}\right]^{\frac{1}{1-\rho}}, \qquad P_{T_i} = \left[\alpha_D\, p_{ii}^{1-\eta} + (1-\alpha_D)\, P_{M_i}^{1-\eta}\right]^{\frac{1}{1-\eta}}, \qquad P_i = \kappa_i\, P_{T_i}^{\alpha_T}\, P_{N_i}^{1-\alpha_T}. \tag{6}$$

Expenditure shares of income are given by:

$$s_{ii} = \alpha_T\, \alpha_D \left(\frac{p_{ii}}{P_{T_i}}\right)^{1-\eta}, \qquad s_{ij} = \alpha_T\,(1-\alpha_D) \left(\frac{P_{M_i}}{P_{T_i}}\right)^{1-\eta} \beta_{ij} \left(\frac{p_{ij}}{P_{M_i}}\right)^{1-\rho}, \quad j \neq i, \tag{7}$$

therefore, $s_{ij} I_i$ is tariff-inclusive spending at consumer prices, $s_{ij} I_i/(1+\tau_{ij})$ is the tariff-exclusive border value. Revenue is rebated lump-sum,

$$I_i = w_i L_i + \sum_{j\neq i} \frac{\tau_{ij}}{1+\tau_{ij}}\, s_{ij} I_i \quad \Longrightarrow \quad I_i = \frac{w_i L_i}{1 - \sum_{j\neq i} \frac{\tau_{ij}}{1+\tau_{ij}}\, s_{ij}}. \tag{8}$$

Trade balances at producer prices in a common currency (since the tariff wedge is a domestic transfer):

$$\mathrm{TB}_i = \sum_{k\neq i} f_{ki} - \sum_{j\neq i} f_{ij}, \qquad f_{ij} \equiv \frac{s_{ij}\, I_i}{1+\tau_{ij}}. \tag{9}$$

By Walras' law only $N-1$ conditions are independent. Furthermore, we impose closure, i.e., no internationally traded assets: $\mathrm{TB}_i = 0$ for all i. Therefore, given $\{\tau_{ij}\}$, the $N-1$ free rates solve $\mathrm{TB}_i = 0$, $i \neq A$, in equilibrium.

2.3 Exchange-rate concepts

Define e_{ij} = units of i's currency per unit of j's, $e_{ii} = 1$, $e_{ij} = e_{ik}e_{kj}$. Therefore, a rise in e_{ij} is a depreciation of i against j. Log perturbations in bilateral exchange rates relative to A are defined by $\nu_j \equiv d\log e_{Aj}$, $\nu_A = 0$. Define relative wages, terms of trade, real exchange rates, and effective exchange rates:

$$\omega_{ij} \equiv \frac{e_{ij}w_j}{w_i}, \qquad \mathrm{ToT}_{ij} \equiv \frac{P_{T_i}}{e_{ij}P_{T_j}} = \frac{1}{e_{ij}}, \qquad q_{ij} \equiv \frac{e_{ij}P_j}{P_i}, \tag{10}$$

$$d\nu_i^E \equiv \sum_{j\neq i} w_{ij}\, d\log e_{ij}, \qquad dq_i^E \equiv \sum_{j\neq i} w_{ij}\, d\log q_{ij}, \qquad w_{ij} = \frac{f_{ij}+f_{ji}}{\sum_{k\neq i}(f_{ik}+f_{ki})}, \tag{11}$$

with $w_{ij} = 1/(N-1)$ in the symmetric model. Under the production normalization,

$$d\log e_{ij} = d\log \omega_{ij} = -\, d\log \mathrm{ToT}_{ij}, \tag{12}$$

so every nominal exchange-rate result is equivalently a relative-wage or terms-of-trade result.

3 General Results on Exchange-Rate Adjustment

Exchange-rate determination under the trade-balance closure has a structure that does not depend on the nested-CES specification.

3.1 The local disturbance–adjustment representation

Consider any differentiable multilateral trade environment with $N-1$ independent trade-balance conditions, $\mathrm{TB}_i = 0$ for $i \neq A$, as functions of the exchange rates and the tariffs. Stacking $d\nu \equiv (\nu_j)_{j\neq A}$ and $d\boldsymbol{\tau} \equiv (dt_{jk})_{j\neq k}$, the conditions for $i \neq A$ are

$$J\, d\nu + G\, d\boldsymbol{\tau} = 0, \qquad J_{il} \equiv \frac{\partial\, d\mathrm{TB}_i}{\partial \nu_l}, \qquad G_{i,(jk)} \equiv \left.\frac{\partial\, d\mathrm{TB}_i}{\partial\, dt_{jk}}\right|_{\nu \text{ fixed}}. \tag{13}$$

Separating the tariff perturbation into a direction a scaled by $d\tau$ (unilateral tariff: $a_{AB} = 1$; war: $a_{AB} = a_{BA} = 1$) gives the impact vector

$$F \equiv Ga \qquad \Longrightarrow \qquad J\, d\nu = -F\, d\tau, \qquad \frac{d\nu}{d\tau} = (-J)^{-1}F. \tag{14}$$

F is the vector of trade-balance disturbances created by the tariff at unchanged exchange rates (incorporating multilateral substitution and income effects), J describes how trade balances respond to exchange rates, and configurations differ only in the direction a applied to the same G.

3.2 The multilateral Marshall–Lerner condition

Let $\widetilde{J}$ be the full $N \times N$ Jacobian before dropping country A. Homogeneity (trade balances depend only on relative currency values) and Walras' law give the identities

$$\sum_l \widetilde{J}_{il} = 0 \quad \forall i, \qquad\qquad \sum_i \widetilde{J}_{il} = 0 \quad \forall l. \tag{15}$$

Assumption 1 (Gross substitutability)**.** *$\widetilde{J}_{il} \geq 0$ for all $l \neq i$. In the symmetric systems below every off-diagonal entry is a positive multiple of D_N, so the assumption there is $D_N > 0$, which holds for $\eta \geq 1$.*

Homogeneity and Assumption 1 give $\widetilde{J}_{ii} = -\sum_{l \neq i} \widetilde{J}_{il} \leq 0$ (i.e., the bilateral Marshall–Lerner conditions), and the reduced J inherits a dominant diagonal,

$$|J_{ii}| = \sum_{l \neq i,\, A} J_{il} + \widetilde{J}_{iA} \;\geq\; \sum_{l \neq i,\, A} J_{il}, \tag{16}$$

strict in every row with $\widetilde{J}_{iA} > 0$. By the standard characterization of irreducibly diagonally dominant Z-matrices:

Result 3.1 (Multilateral Marshall–Lerner condition)**.** *Under Assumption 1, with J irreducible and $\widetilde{J}_{iA} > 0$ for some i, $-J$ is a nonsingular M-matrix:*

$$\det(-J) > 0, \qquad (-J)^{-1} \geq 0 \textit{ elementwise}, \qquad \operatorname{Re} \lambda(J) < 0. \tag{17}$$

In words, the multilateral Marshall–Lerner condition generalizes the bilateral Marshall–Lerner condition to require that the exchange-rate adjustment mechanism is regular in the sense that each country's own depreciation improves its own trade balance, no country's appreciation worsens another country's balance, and the own effect dominates the cross effects. Therefore, any pattern of trade imbalances is eliminated by "conventional" exchange rate adjustment: surplus countries appreciate, deficit countries depreciate.[1]

Result 3.2 (Sign rule)**.** *Under* (17)*,*

$$\operatorname{sign}\!\left(\frac{d\nu_i}{d\tau}\right) = \operatorname{sign}\!\left(\sum_j \omega_{ij} F_j\right), \qquad \omega_{ij} \equiv \left[(-J)^{-1}\right]_{ij} \geq 0. \tag{18}$$

Under the multilateral Marshall–Lerner condition, opposite-signed bilateral responses require a mixed-sign F, i.e., bilateral exchange-rate ambiguity comes from the tariff disturbance on trade balances and not from the adjustment mechanism.

[1]Formally: gross substitutability $\Rightarrow$ multilateral Marshall–Lerner. We assume the former as the primitive (Assumption 1), but the subsequent analysis uses only (17). Under the sign structure of Assumption 1, multilateral Marshall–Lerner $\equiv$ local stability of the tatonnement $\dot{\nu}_i = k_i \mathrm{TB}_i$, any $k_i > 0$, which says that simple dynamic exchange-rate adjustment processes converge to the equilibrium.

4 Tariff Responses in the Model

This section computes the entries of J and G for the nested-CES model for $N = 2$, $N = 3$, and general symmetric N.

4.1 The linearized system

Linearizing at a balanced free-trade baseline with normalized incomes, define home share $h_i = \alpha_D$, within-import shares $b_{ij} = \beta_{ij}$, income shares $s_{ij} = m\, b_{ij}$ with $m \equiv \alpha_T(1-\alpha_D)$, and flows $f_{ij} = s_{ij} I_i$, $\sum_k f_{ki} = \sum_j f_{ij}$. Perturbing this baseline with tariffs dt_{ij}:

$$dp_{ij} = \nu_j - \nu_i + dt_{ij} \qquad (dp_{ii} = 0), \tag{19}$$

$$d\log P_{M_i} = \sum_{j\neq i} b_{ij}\, dp_{ij}, \qquad d\log P_{T_i} = (1-\alpha_D)\, d\log P_{M_i}, \tag{20}$$

$$d\log s_{ii} = -(1-\eta)(1-\alpha_D)\, d\log P_{M_i}, \tag{21}$$

$$d\log s_{ij} = \underbrace{(1-\rho)\big(dp_{ij} - d\log P_{M_i}\big)}_{\text{across origins}} + \underbrace{(1-\eta)\,\alpha_D\, d\log P_{M_i}}_{\text{home vs. imports}}, \tag{22}$$

$$d\log I_i = \nu_i + \sum_{j\neq i} s_{ij}\, dt_{ij}, \tag{23}$$

$$d\,\mathrm{TB}_i = \sum_{k\neq i} f_{ki}\big(d\log s_{ki} + d\log I_k - dt_{ki}\big) - \sum_{j\neq i} f_{ij}\big(d\log s_{ij} + d\log I_i - dt_{ij}\big). \tag{24}$$

Reading the coefficients off (24) gives the model's entries of J and G in (13), so we obtain specific parametric expressions for every term from the model of Section 2.

The algebraic structure of (14) also bounds how bilateral signs can change with ρ. The elasticity ρ enters the share responses (22) only through the coefficient $(1-\rho)$, so every entry of J and F is linear in ρ. Each entry of $\mathrm{adj}(-J)$ is a determinant of an $(N-2)\times(N-2)$ submatrix of $-J$ and is therefore a polynomial of degree at most $N-2$ in ρ, so each numerator $[\mathrm{adj}(-J)\,F]_i$ in (14) is a polynomial of degree at most $N-1$. Since $\det(-J) > 0$ under (17), a bilateral response can change sign only at a root of its numerator polynomial. At $N = 3$ the symmetric baseline reduces the relevant numerator to a linear function of ρ, giving the single threshold derived below; a general asymmetric baseline gives a quadratic (Appendix B); and at $N \geq 4$ the higher-degree numerators allow a response to change sign more than once as ρ rises.

Now suppose A imposes $d\tau = dt_{AB}$ at the symmetric baseline. Define the following constant, a function of model parameters alone:

$$\rho_1^* \equiv 1 + \alpha_D(\eta - 1) - \alpha_T(1-\alpha_D) = \underbrace{\alpha_D\,\eta}_{\text{home switching}} + \underbrace{(1-\alpha_D)(1-\alpha_T)}_{\text{expenditure absorption}} \;>\; 0. \tag{25}$$

4.2 Two countries

At $N = 2$ the system (14) is scalar and ρ is inoperative ($dp_{AB} = d\log P_{M_A}$). With $s_{AB} = s_{BA} = m$, equations (19)–(24) collect to

$$d\,\mathrm{TB}_A = \underbrace{m\,D_2\,\nu_B}_{\text{adjustment}} + \underbrace{m\,\rho_1^*\,d\tau}_{\text{impact disturbance}} = 0, \qquad D_2 \equiv 2\alpha_D(\eta - 1) + 1, \tag{26}$$

$$\boxed{\left.\frac{d\log e_{AB}}{d\tau}\right|_{\tau=0} = -\frac{\rho_1^*}{D_2}.} \tag{27}$$

In reduced form, with export volume $X(1/e)$, import volume $M((1+\tau)e)$, and elasticities $\varepsilon_X, \varepsilon_M$, balanced trade gives $\partial \mathrm{TB}_A/\partial e = M(\varepsilon_X + \varepsilon_M - 1)$; comparing with (26) per unit of import value,

$$\varepsilon_X + \varepsilon_M - 1 = D_2, \qquad \varepsilon_X = \varepsilon_M = \alpha_D(\eta - 1) + 1. \tag{28}$$

Result 4.1 (Conventional wisdom, $N = 2$)**.** *At $N = 2$, condition* (17) *collapses to $D_2 > 0$, which is exactly the textbook Marshall–Lerner condition $\varepsilon_X + \varepsilon_M > 1$; in the nested-CES model it holds whenever $\eta \geq 1$. The impact disturbance is $F = m\rho_1^* > 0$ for all admissible parameters. A unilateral import tariff therefore appreciates the tariffing country's currency: $d\log e_{AB}/d\tau = -\rho_1^*/D_2 < 0$.*

The two ingredients of the general sign rule appear separately here: $D_2 > 0$ is the $N = 2$ analog of the multilateral Marshall–Lerner condition (17), and $\rho_1^* > 0$ is the analog of the condition on F. At $N = 2$ the disturbance is a scalar with an unambiguous sign — at unchanged exchange rates the tariff improves A's balance by $m\rho_1^*\,d\tau$, through home switching and non-tradable absorption — and hence the exchange rate appreciates in equilibrium to eliminate the surplus, with no threshold condition. Bilateral ambiguity can first arise at $N = 3$, where F acquires a second component. (Appendix A gives the exact finite-tariff Cobb–Douglas solution, which confirms the sign globally.)

4.3 Three countries

At $N = 3$ ($\beta_{ij} = \frac{1}{2}$, $s_{ij} = m/2$), (19)–(20) give

$$d\log P_{M_A} = \tfrac{1}{2}(\nu_B + \nu_C + d\tau), \qquad d\log P_{M_B} = \tfrac{1}{2}\nu_C - \nu_B, \qquad d\log P_{M_C} = \tfrac{1}{2}\nu_B - \nu_C, \tag{29}$$

with $d\log I_A = \frac{m}{2}d\tau$, $d\log I_B = \nu_B$, $d\log I_C = \nu_C$. Substituting into (24) for $i = B, C$ and collecting:

$$J = -\frac{mD_3}{4}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \qquad F = -\frac{m}{4}\begin{pmatrix} \rho + \rho_1^* \\ \rho_1^* - \rho \end{pmatrix}, \qquad D_3 = 3\alpha_D(\eta - 1) + \rho + 1. \tag{30}$$

The impact disturbances are $F_B = -\frac{m}{4}(\rho + \rho_1^*) < 0$ and $F_C = \frac{m}{4}(\rho - \rho_1^*)$: the target always loses; the bystander gains diverted expenditure (increasing in ρ) against reduced sales to a poorer B and

A's home switching, with net gain iff $\rho > \rho_1^*$. Inverting:

$$\begin{pmatrix} \nu_B \\ \nu_C \end{pmatrix} = (-J)^{-1} F \, d\tau = -\frac{1}{3D_3} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} \rho + \rho_1^* \\ \rho_1^* - \rho \end{pmatrix} d\tau = \frac{1}{3D_3} \begin{pmatrix} -(\rho + 3\rho_1^*) \\ \rho - 3\rho_1^* \end{pmatrix} d\tau. \tag{31}$$

Result 4.2 (Three-country threshold)**.** *With $D_3 > 0$,*

$$\frac{d \log e_{AB}}{d\tau} = -\frac{\rho + 3\rho_1^*}{3D_3} < 0, \qquad \boxed{\frac{d \log e_{AC}}{d\tau} = \frac{\rho - 3\rho_1^*}{3D_3}} \qquad \frac{d \log e_{BC}}{d\tau} = \frac{2\rho}{3D_3} > 0 : \tag{32}$$

A always appreciates against B, B always depreciates against C, and A depreciates against the untariffed C iff $\rho > \rho_3^* \equiv 3\rho_1^*$.

The distinction between the two thresholds deserves emphasis. A favorable trade-diversion effect on C at unchanged exchange rates ($F_C > 0$, i.e. $\rho > \rho_1^*$) is not sufficient for the A–C bilateral rate to reverse. By the sign rule, $d\nu_C/d\tau \propto F_B + 2F_C \propto \rho - 3\rho_1^*$: C's equilibrium rate responds to the entire disturbance vector through $(-J)^{-1}F$, mixing C's own diversion gain with its exposure to B's deterioration, and reversal requires the gain to outweigh that exposure. The gap between the impact threshold ρ_1^* and the equilibrium threshold $3\rho_1^*$ is thus a genuinely multilateral distinction between the impact incidence of the tariff and the equilibrium relative-price adjustment.

Corollary (single elasticity). With $\rho = \eta = \sigma$: $\rho_3^* - \sigma = \sigma(3\alpha_D - 1) + 3(1 - \alpha_D)(1 - \alpha_T) > 0$ whenever $\alpha_D \geq 1/3$, for any α_T; the reversal is unreachable. At the equal-weights point $\alpha_D = 1/3$, $\alpha_T = 3/4$: $d \log e_{AC}/d\tau = -1/(12\sigma) \to 0^-$.

4.4 General symmetric N

With weights $1/(N-1)$ and the $N-2$ bystanders identical (common ν_C), (19)–(20) give

$$d \log P_{M_A} = \frac{\nu_B + d\tau + (N-2)\nu_C}{N-1}, \qquad d \log P_{M_B} = \frac{(N-2)\nu_C}{N-1} - \nu_B, \qquad d \log P_{M_C} = \frac{\nu_B - 2\nu_C}{N-1}, \tag{33}$$

with $d \log I_A = \frac{m}{N-1} d\tau$, $d \log I_B = \nu_B$, $d \log I_C = \nu_C$. Collecting (24) for B and a representative bystander (rows scaled by $(N-1)/m$):

$$D_N \begin{pmatrix} -(N-1) & N-2 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} \nu_B \\ \nu_C \end{pmatrix} = \begin{pmatrix} (N-2)\rho + \rho_1^* \\ \rho_1^* - \rho \end{pmatrix} d\tau, \qquad D_N = N\alpha_D(\eta - 1) + (N-2)\rho + 1, \tag{34}$$

where the counting matrix has determinant N; inverting:

$$\frac{d \log e_{AB}}{d\tau} = -\frac{N\rho_1^* + (N-2)\rho}{N\,D_N} < 0, \qquad \frac{d \log e_{AC}}{d\tau} = \frac{\rho - N\rho_1^*}{N\,D_N}, \qquad \frac{d \log e_{BC}}{d\tau} = \frac{(N-1)\,\rho}{N\,D_N} > 0, \tag{35}$$

which reduce to (32) at $N = 3$ and to (27) at $N = 2$.

Result 4.3 (Dilution threshold)**.** *Under* (17)*, the bystander bilateral rate reverses iff*

$$\boxed{\rho > \rho_N^* = N\rho_1^*.} \tag{36}$$

Diverted expenditure is spread across $N-2$ bystanders, each with weight $1/(N-1)$, so stronger cross-origin substitution is required to reverse any individual bilateral rate (dilution). The exact linearity in N is a symmetric nested-CES result, and away from symmetry no universal threshold exists; the reversal region itself is not a symmetry artifact, however — at any balanced baseline where A and a bystander both import from both foreign origins, sufficiently strong cross-origin substitution still reverses the bystander rate (Appendix B). With $\rho = \eta = \sigma$, $\rho_N^* - \sigma = \sigma(N\alpha_D - 1) + N(1-\alpha_D)(1-\alpha_T) > 0$ whenever $\alpha_D \geq 1/N$: the reversal stays unreachable for single-elasticity models on the home-biased region.

5 Bilateral versus Effective Exchange Rates

Averaging (35) with symmetric weights:

$$\frac{d\nu_A^E}{d\tau} = \frac{\nu_B + (N-2)\nu_C}{N-1} = \boxed{-\frac{\rho_1^*}{D_N}} < 0, \qquad \nu_B - \nu_C = -\frac{(N-1)\rho}{ND_N}\, d\tau. \tag{37}$$

Result 5.1 (NEER preservation)**.** *Under symmetry, a unilateral tariff appreciates the tariffing country's NEER for every $N \geq 2$ and every ρ satisfying* (17)*, even when $\rho > N\rho_1^*$ reverses the bilateral rate against every bystander.*

ρ governs the relative distribution of adjustment across foreign partners: its direct terms cancel from the NEER numerator under symmetry, while it remains in D_N and therefore still affects the magnitude of the aggregate response. The two-country appreciation result thus survives multilateralization as an effective, rather than universally bilateral, statement — it is bilateral at $N = 2$ only because there is one foreign country. The exact cancellation is specific to the symmetric nested-CES environment; the distinction between aggregate home-versus-foreign adjustment and foreign-supplier reallocation is more general. Appendix B quantifies the robustness of the cancellation: within a structured asymmetric family it is preserved to leading order in $1/N$, and it can reverse only for a disproportionately under-weighted target at ρ of order N. (Appendix C collects the correspondence across N.)

6 Extensions

6.1 Alternative tariff configurations

Responses are linear in the tariff vector, so configurations are superpositions of directed bilateral shocks (directions a on the same G).

Broad unilateral tariff. A uniform tariff on all $N-1$ partners treats every origin identically, deactivating cross-origin reallocation. Superposing (35) over targets:

$$\frac{d\log e_{Aj}}{d\tau} = -\frac{(N-1)\rho_1^*}{D_N} = \frac{d\nu_A^E}{d\tau} \qquad \forall j : \tag{38}$$

ρ drops out of the numerator, every bilateral rate appreciates equally, and bilateral and effective responses coincide. Bilateral ambiguity arises from discrimination across foreign suppliers, not protection itself.

Symmetric trade war. Adding the mirror shock $dt_{BA} = d\tau$:

$$\left.\frac{d\log e_{AB}}{d\tau}\right|^{w} = 0, \qquad \left.\frac{d\log e_{AC}}{d\tau}\right|^{w} = \frac{\rho - \rho_1^*}{D_N}, \qquad \left.\frac{d\nu_A^E}{d\tau}\right|^{w} = \frac{N-2}{N-1}\,\frac{\rho-\rho_1^*}{D_N}. \tag{39}$$

Result 6.1 (War threshold)**.** *Under* (17)*, in a symmetric war the belligerents' effective rates depreciate, and every bystander appreciates against both, iff $\rho > \rho_1^*$, independently of N.*

Retaliation cancels the direct bilateral component between A and B, leaving the bystander reallocation margin decisive: each bystander's equation decouples, so its own impact effect sets the sign and does not dilute with N. The war threshold is far below the unilateral threshold $N\rho_1^*$ because the direct appreciation against B has been removed. At $\eta = 1$, $\rho_1^* = 1 - m < 1$.

6.2 Real exchange rates

From (10) and (6), $d\log q_{Aj} = \nu_j + \alpha_T(d\log P_{T_j}^{\text{own}} - d\log P_{T_A}^{\text{own}})$; for the unilateral tariff both bilateral real rates take one form at every N:

$$d\log q_{Aj} = \left(1 - \frac{mN}{N-1}\right)\nu_j \;-\; \frac{m}{N-1}\,d\tau, \qquad j = B, C: \tag{40}$$

attenuated nominal pass-through plus a direct price-index term working toward real appreciation of A. Setting $d\log q_{AC} = 0$ with (35) (for $m < 1/N$):

$$\rho_q^*(N) = \frac{\big[(N-1) - mN\big]\,N\rho_1^* + mN\big[N\alpha_D(\eta-1)+1\big]}{(N-1)(1-mN)} \;>\; N\rho_1^*, \tag{41}$$

Averaging (40):

$$\frac{dq_A^E}{d\tau} = -\left(1 - \frac{mN}{N-1}\right)\frac{\rho_1^*}{D_N} - \frac{m}{N-1} \;<\; 0 \qquad \text{for all } \rho, N \;\left(m < \tfrac{N-1}{N}\right). \tag{42}$$

Result 6.2 (Real rates)**.** *Under* (17)*, the direct CPI effect makes real bilateral reversal harder than nominal reversal, $\rho_q^*(N) > N\rho_1^*$, and reinforces effective appreciation: the REER response is negative for all ρ and N, and strictly larger in magnitude than the pass-through of the nominal response alone.*

7 Relation to the Canonical PTA Model

7.1 Competing exporters

The canonical benchmark is the competing-exporter environment of the preferential-trade-agreement literature, in the tradition of Viner (1950) and Mundell (1964). Importer A holds only the numeraire

y; exporters B, C hold good x; preferences are quasilinear, $U_i = u_i(x_i) + y_i$. A levies specific tariffs t_B, t_C; with q_j the exporter price and p the domestic price,

$$p = q_B + t_B = q_C + t_C, \qquad M_A(p) = E_B(p - t_B) + E_C(p - t_C), \qquad M_A' < 0,\ E_j' > 0, \tag{43}$$

where $M_A(p) = x_A^d(p)$ and $E_j(q_j) = \omega_j - x_j^d(q_j)$ from $u_i' =$ local price. Differentiating in t_B:

$$\frac{dp}{dt_B} = \frac{E_B'}{E_B' + E_C' - M_A'} \in (0,1), \qquad \frac{dq_B}{dt_B} = -\frac{E_C' - M_A'}{E_B' + E_C' - M_A'} < 0, \qquad \frac{dq_C}{dt_B} = \frac{dp}{dt_B} > 0. \tag{44}$$

Result 7.1 (Viner–Mundell)**.** *A discriminatory tariff on B worsens B's and improves C's terms of trade, unconditionally.*

7.2 Relation to the present model

An improvement in C's terms of trade is a rise in e_{AC}; the nested model requires $\rho > 3\rho_1^*$, the canonical model nothing. The canonical environment removes the margins that oppose diversion toward C. Decomposing the threshold:

$$\rho_3^* = 3\rho_1^* = \underbrace{3}_{B\text{–}C\text{ linkage}} + \underbrace{3\alpha_D(\eta - 1)}_{\text{home switching}} - \underbrace{3\alpha_T(1 - \alpha_D)}_{\text{openness credit}}. \tag{45}$$

The linkage term collects the forces running through B–C trade: B's income loss cuts its purchases from C, and relative prices induce C to substitute toward B's cheapened goods. The second term is A's switching toward its own tradable; the openness credit reflects that a larger import share m gives the tariff more foreign expenditure to reallocate. Each piece is an identifiable economic force, so the threshold is not a CES-curvature artifact. The canonical model deletes the first two opposing forces and the income effects behind the first: B and C are competing exporters rather than full trading partners (no B–C demand linkage), A does not produce the imported good (no home switching), and quasilinearity removes income effects. What remains is reallocation between the two foreign suppliers, signed directly by $M_A' < 0$ and $E_j' > 0$.

Result 7.2. *The canonical PTA model is the restriction of the three-country model to a subspace on which $\rho_3^* \leq 0$: the unconditional Viner–Mundell sign is the threshold condition evaluated where it cannot bind. The nested model embeds the same diversion mechanism among the opposing margins, and $\rho > \rho_3^*$ states when it dominates.*

This locates the analytical trade-agreement literature by model class rather than by conclusion. Competing-exporter models of preferential agreements (Bagwell and Staiger, 1999; Ornelas, 2005) study exactly the canonical environment: B and C sell to A but do not trade with each other, and quasilinearity removes income effects, so the Viner–Mundell sign holds unconditionally. Symmetric bloc and endowment structures (Krugman, 1991; Bond and Syropoulos, 1996) reach the same sharpness along the other route, with little or no home production of importables; Viner's (1950) original identical-goods case is the limit $\rho = \infty$. Kemp and Wan (1976) illustrate the converse: with none of these restrictions imposed, no unconditional sign is claimed. Applied models sit at the opposite pole: GTAP-class and modern quantitative trade models retain every margin in (45) and generate either sign in simulations, without an explicit analytical threshold. The condition $\rho > 3\rho_1^*$ is that threshold.[2]

8 Conclusion

The general equilibrium response to a tariff is $(-J)^{-1}F$: the tariff creates trade-balance disturbances at unchanged exchange rates, and the adjustment mechanism, regular under the multilateral Marshall–Lerner condition, combines them with nonnegative weights. At $N = 2$ this reduces to the textbook Marshall–Lerner appreciation result. At $N \geq 3$, a discriminatory tariff generates foreign-supplier reallocation and hence a mixed-sign disturbance vector: the tariffing country always appreciates against the target, but under symmetry depreciates against a bystander iff $\rho > N\rho_1^*$, a threshold that rises with N because diversion is diluted across bystanders. Aggregation removes the direct reallocation term from the NEER numerator, so the effective exchange rate appreciates for every ρ and N; configuration and asymmetry set the limits of this aggregate result, with reciprocal retaliation reversing it at $\rho > \rho_1^*$ and, as Appendix B shows, an under-weighted target reversing it at ρ of order N.

The organizing distinction is between aggregate home-versus-foreign adjustment and reallocation across foreign suppliers. Under symmetry, the aggregate margin determines the sign of the effective response, while reallocation determines its bilateral distribution; away from symmetry, reallocation also affects the magnitude of the aggregate response and, for a sufficiently under-weighted target, its sign (Appendix B). Tariffs change both how much a country buys from abroad and from whom it buys it.

References

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Ornelas, Emanuel (2005). "Trade Creating Free Trade Areas and the Undermining of Multilateralism." *European Economic Review* 49(7), 1717–1735.

[2] Severing the B–C linkage alone accounts for most of the threshold, and combining it with $\alpha_D = 0$ removes the reversal region entirely.

Viner, Jacob (1950). *The Customs Union Issue.* New York: Carnegie Endowment for International Peace.

A Exact Cobb–Douglas solution ($N = 2$)

At $\eta = 1$ shares are constant at the weights, $I_A = w_A L_A/[1 - \frac{\tau}{1+\tau} m]$ from (8), and $\mathrm{TB}_A = 0$ reads $e\, m\, w_B L_B = m I_A/(1+\tau)$:

$$e(\tau) = \frac{w_A L_A}{w_B L_B} \cdot \frac{1}{1+(1-m)\tau}, \qquad \left.\frac{d \log e}{d\tau}\right|_{\tau=0} = -(1-m) = -\left.\frac{\rho_1^*}{D_2}\right|_{\eta=1}, \tag{46}$$

globally decreasing in τ: the linearized sign in (27) is exact at all tariff levels in this case. ($\eta = 1$ fixes the elasticity, not the weight α_D.)

B Asymmetry

The disturbance-adjustment representation (14) and the multilateral Marshall–Lerner condition (17) do not use symmetry; the exact thresholds of Section 4 and the NEER cancellation of Section 5 do. This appendix establishes what survives without it: the bilateral reversal region survives at any interior baseline, and the NEER result is preserved to leading order within a structured asymmetric family.

General asymmetric baseline. At any balanced baseline, (19)–(24) hold with observed shares (b_{ij}, h_i) and incomes y_i as coefficients. For $N = 3$ at a general baseline (paper appendix), the bystander slope has denominator $\det(-J) > 0$ and numerator quadratic in ρ,

$$\operatorname{sign} \frac{d \log e_{AC}}{d\tau} = \operatorname{sign}\big(c_2\, \rho^2 + c_1\, \rho + c_0\big), \qquad c_2 = b_{AB}\, b_{AC}\, b_{CA}\, b_{CB}\, (1-h_A)(1-h_C)\, y_A\, y_C, \tag{47}$$

with c_1, c_0 depending on the full pattern of shares and sizes. Since $c_2 > 0$ whenever A and C each buy from both origins, sufficiently strong cross-origin substitution reverses e_{AC} at any such baseline; the threshold is the largest root, equal to $3\rho_1^*$ at the symmetric point. Away from symmetry no universal bilateral threshold exists.

Structured asymmetry. Let the target carry a multiple κ of the symmetric bilateral share (comparable across N, since the symmetric share shrinks like $1/(N-1)$), with the weight matrix symmetric and balanced:

$$b_{AB} = b_{BA} = \frac{\kappa}{N-1}, \qquad b_{AC} = b_{CA} = b_{BC} = b_{CB} = \frac{1 - \frac{\kappa}{N-1}}{N-2}, \qquad b_{CC'} = \frac{1 - 2\, b_{CA}}{N-3}. \tag{48}$$

The reduced system again has two unknowns and solves in closed form. With NEER weights $w_{Aj} = b_{Aj}$, the response is a rational function of (ρ, N, κ) collapsing to $-\rho_1^*/D_N$ at $\kappa = 1$, with numerator $-\kappa\rho_1^*$ times a quadratic $Q(\rho)$ whose leading coefficient is proportional to $(\kappa-1)(N-1-\kappa)$.

Result B.1 (Asymmetric NEER). *Holding κ fixed as $N \to \infty$,*

$$N \, \frac{d\nu_A^E}{d\tau} \;\longrightarrow\; -\,\kappa \, \frac{\rho_1^*}{\rho + \alpha_D(\eta - 1)} = -\,\kappa \lim_{N\to\infty} \frac{N\rho_1^*}{D_N}. \tag{49}$$

To leading order, asymmetric exposure scales the symmetric NEER response by the target's relative importance and preserves its sign. At finite N: for $\kappa \geq 1$ all coefficients of Q are positive at large N, so effective appreciation holds for every ρ; for $\kappa < 1$, Q has a single positive root $\rho_E^*(\kappa, N) = \frac{N\rho_1^*}{1-\kappa}(1 + O(N^{-1}))$, beyond which the NEER depreciates — an under-weighted target lets depreciations against many bystanders outweigh the appreciation against B, with $\rho_E^* \to N\rho_1^*$ as $\kappa \to 0$. The reversal threshold is of order N. The symmetric NEER result is exact under symmetry and approximately robust within this structured asymmetric family; robustness to arbitrary asymmetric trade networks is not claimed.

C Correspondence across N

object	$N = 2$	$N = 3$	general N
ML condition (17)	$\varepsilon_X + \varepsilon_M - 1 = D_2 > 0$	$D_3 > 0$	$-J$ a nonsingular M-matrix
sufficient primitive	$\eta \geq 1$	$\eta \geq 1$	gross substitutability (Assumption 1)
sign rule	one impact, unconditional	weights $\frac{1}{3}\left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$	$\operatorname{sign}\left(\sum_j \omega_{ij} F_j\right)$, $\omega \geq 0$
bilateral threshold	none (no bystander)	$3\rho_1^*$	$N\rho_1^*$ (symmetric); degree-$(N-1)$ roots in general
real bilateral threshold	none	$\rho_q^*(3)$	$\rho_q^*(N)$ (41)
NEER (unilateral)	$-\rho_1^*/D_2$	$-\rho_1^*/D_3$	$-\rho_1^*/D_N$: sign invariant under symmetry
war threshold	—	ρ_1^*	ρ_1^*, independent of N